Answers: Solving first-order linear ODEs using integrating factors

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Summary

Answers to questions relating to the guide on solving first-order linear ODEs using integrating factors.

These are the answers to Questions: Introduction to solving first-order linear ODEs using integrating factors.

Please attempt the questions before reading these answers!

Q1

1.1. The equation rearranged into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ is $\frac{\mathrm{d}y}{\mathrm{d}x} + y = -8$.

1.2. The equation rearranged into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ is $\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{2x} \cdot y = -5x$.

1.3. The equation rearranged into the form $\frac{\mathrm{d}x}{\mathrm{d}t} + P(t)x = Q(t)$ is $\frac{\mathrm{d}x}{\mathrm{d}t} + \cos(t)x = \cos(t)$.

1.4. The equation rearranged into the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ is $\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{1}{x} \cdot y = x$.

Q2

2.1. The integrating factor for $-\frac{\mathrm{d}y}{\mathrm{d}x} = y + 8$ is $\mu = e^{x}$.

2.2. The integrating factor for $\frac{1}{2x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}x} + 5x}{y}$ is $\mu = x^{-\frac{1}{2}}$.

2.3. The integrating factor for $\frac{\mathrm{d}x}{\mathrm{d}t} - \cos(t) = -x\cos(t)$ is $\mu = e^{\sin(t)}$.

2.4. The integrating factor for $x\frac{\mathrm{d}y}{\mathrm{d}x} - x^2 = y$ is $\mu = x^{-1}$.

Q3

3.1. The general solution is $y = -8 + Ce^{-x}$.

3.2. The general solution is $y = -\frac{10}{3}x^2 + Cx^{\frac{1}{2}}$.

3.3. The general solution is $x = 1 + Ce^{-\sin(t)}$.

3.4. The general solution is $y = x^2 + Cx$.

Q4

4.1. The exact solution, given the initial condition $y(0) = 0$, is $y = -8 + 8e^{-x}$.

4.2. The exact solution, given the initial condition $y(1) = 6$, is $y = -\frac{10}{3}x^2 + \frac{28}{3}x^{\frac{1}{2}}$.

4.3. The exact solution, given the initial condition $x(0) = \pi$, is $x = 1 + (\pi - 1)e^{-\sin(t)}$.

4.4. The exact solution, given the initial condition $y(2) = 3$, is $y = x^2 - \frac{1}{2}x$.

Q5

5.1. The general solution is $y = -x^2 - \frac{1}{2}x + Cx^5$

5.2 The general solution is $y = \frac{\frac{10}{3}x^6 + x^5 + C}{4x+1}$

5.3 The exact solution, given the initial condition $y(\frac{\pi}{2}) = 1$, is $y(x) = \frac{1}{5}\sin(x) + \frac{4}{5}\csc^4(x)$

5.4 The exact solution, given the initial condition $y(0) = 4$, is $y(x) = -4xe^{-4x}\cos(x) + 4e^{-4x}\sin(x) + 4e^{-4x}$

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